

1 Relations and Functions - Concepts & Formulas

Chapter Concepts

A relation R in a set A is called:

- **Reflexive** if $(a, a) \in R$ for every $a \in A$
- **Symmetric** if $(a, b) \in R \Rightarrow (b, a) \in R$ for every $a, b \in A$
- **Transitive** if $(a, b) \in R$ and $(b, c) \in R \Rightarrow (a, c) \in R$ for every $a, b, c \in A$

A relation is an **equivalence relation** if it is reflexive, symmetric and transitive.

A function $f : X \subset \mathbb{R} \rightarrow Y \subset \mathbb{R}$ is:

- **One-one (injective)** if the images of distinct elements of X under f are distinct
- **Onto (surjective)** if every element of Y is the image of some element of X
- **Bijjective** if it is both one-one and onto

Key Formulas & Properties

- Number of relations from A to B : 2^{mn} where $|A|=m, |B|=n$
- Number of functions from A to B : n^m
- Number of one-one functions from A to B : ${}^n P_m$ if $n \geq m$, else 0
- Composition of functions: $(f \circ g)(x) = f(g(x))$
- Inverse function exists only for bijective functions

Important Types of Functions

- **Identity function:** $f(x) = x$
- **Constant function:** $f(x) = c$
- **Modulus function:** $f(x) = |x|$
- **Signum function:** $f(x) = \frac{|x|}{x}$ for $x \neq 0$, $f(0) = 0$
- **Greatest integer function:** $f(x) = [x]$

1 Relations and Functions - Practice Questions

Practice Questions (7)

1. Show that $f : \mathbb{R} \rightarrow \mathbb{R}$ defined by $f(x) = \frac{x}{x^2+1} \forall x \in \mathbb{R}$ is neither one-one nor onto.

2. Let R be a relation on $\mathbb{N} \times \mathbb{N}$ defined by $(a, b)R(c, d)$ iff $ad(b+c) = bc(a+d)$. Show that R is an equivalence relation.

3. Show that the function f in $A = \mathbb{R} - \{\frac{2}{3}\}$ defined as $f(x) = \frac{4x+3}{6x-4}$ is one-one and onto. Also if $f(x) = y$, express x in terms of y .

4. Let $A = \{1, 2, 3, \dots, 9\}$ and R be the relation in $A \times A$ defined by $(a, b)R(c, d)$ if $a + d = b + c$ for $(a, b), (c, d)$ in $A \times A$. Prove that R is an equivalence relation.

5. Show that the relation S in the set $A = \{x \in \mathbb{Z} : 0 \leq x \leq 12\}$ given by $S = \{(a, b) : a, b \in \mathbb{Z}, |a - b| \text{ is divisible by } 4\}$ is an equivalence relation.

6. Let $A = \mathbb{R} - \{3\}, B = \mathbb{R} - \{1\}$. If $f : A \rightarrow B$ be defined by $f(x) = \frac{x-2}{x-3}, \forall x \in A$. Show that f is bijective.

7. A relation R is defined on a set of real numbers $\mathbb{R}$ as $R = \{(x, y) : x \cdot y \text{ is an irrational number}\}$. Check whether R is reflexive, symmetric and transitive or not.

2 Inverse Trigonometric Functions - Concepts & Formulas

Chapter Concepts

Inverse trigonometric functions are the inverse functions of trigonometric functions. They are defined with restricted domains to make them one-to-one and hence invertible.

Principal Value Branches:

- $\sin^{-1} x$: Domain $[-1, 1]$, Range $[-\frac{\pi}{2}, \frac{\pi}{2}]$
- $\cos^{-1} x$: Domain $[-1, 1]$, Range $[0, \pi]$
- $\tan^{-1} x$: Domain $\mathbb{R}$, Range $(-\frac{\pi}{2}, \frac{\pi}{2})$
- $\cot^{-1} x$: Domain $\mathbb{R}$, Range $(0, \pi)$
- $\sec^{-1} x$: Domain $\mathbb{R} - (-1, 1)$, Range $[0, \pi] - \{\frac{\pi}{2}\}$
- $\operatorname{cosec}^{-1} x$: Domain $\mathbb{R} - (-1, 1)$, Range $[-\frac{\pi}{2}, \frac{\pi}{2}] - \{0\}$

Key Formulas & Properties

- $\sin^{-1}(-x) = -\sin^{-1} x$
- $\cos^{-1}(-x) = \pi - \cos^{-1} x$
- $\tan^{-1}(-x) = -\tan^{-1} x$
- $\sin^{-1} x + \cos^{-1} x = \frac{\pi}{2}$
- $\tan^{-1} x + \cot^{-1} x = \frac{\pi}{2}$
- $\sec^{-1} x + \operatorname{cosec}^{-1} x = \frac{\pi}{2}$
- $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \left(\frac{x+y}{1-xy} \right)$ if $xy < 1$
- $\tan^{-1} x - \tan^{-1} y = \tan^{-1} \left(\frac{x-y}{1+xy} \right)$ if $xy > -1$

Important Conversions

- $\sin^{-1} x = \cos^{-1} \sqrt{1-x^2} = \tan^{-1} \frac{x}{\sqrt{1-x^2}}$
- $\cos^{-1} x = \sin^{-1} \sqrt{1-x^2} = \tan^{-1} \frac{\sqrt{1-x^2}}{x}$
- $\tan^{-1} x = \sin^{-1} \frac{x}{\sqrt{1+x^2}} = \cos^{-1} \frac{1}{\sqrt{1+x^2}}$
- $2 \tan^{-1} x = \sin^{-1} \frac{2x}{1+x^2} = \cos^{-1} \frac{1-x^2}{1+x^2} = \tan^{-1} \frac{2x}{1-x^2}$

2 Inverse Trigonometric Functions - Practice Questions

Practice Questions (8)

1. Find the value of $\sin^{-1}[\sin(\frac{13\pi}{7})]$.

2. Find the principal value of $\tan^{-1}(\sqrt{3}) - \sec^{-1}(-2)$.

3. Evaluate $\sin[\frac{\pi}{3} - \sin^{-1}(-\frac{\sqrt{3}}{2})]$.

4. Evaluate $\sin^{-1}[\sin(\frac{13\pi}{7})] + \cos^{-1}[\cos(\frac{13\pi}{7})]$.

5. Find the value of $\tan^{-1}[2 \cos(2 \sin^{-1}(\frac{1}{2}))] + \tan^{-1}(1)$.

6. Find the principal value of $\tan^{-1}(1) + \cos^{-1}(-\frac{1}{2}) + \sin^{-1}(-\frac{1}{\sqrt{2}})$.

7. Write the value of $\tan^{-1}[2 \sin(2 \cos^{-1}(\frac{\sqrt{3}}{2}))]$.

8. Evaluate: $\cos^{-1}(\cos(\frac{2\pi}{3})) + \sin^{-1}(\sin(\frac{2\pi}{3}))$.

3 Matrices and Determinants - Concepts & Formulas

Chapter Concepts

A matrix is a rectangular array of numbers arranged in rows and columns. Determinants are scalar values that can be computed from square matrices.

Types of Matrices:

- **Row Matrix:** Matrix with only one row
- **Column Matrix:** Matrix with only one column
- **Square Matrix:** Matrix with equal number of rows and columns
- **Diagonal Matrix:** Square matrix with all non-diagonal elements zero
- **Scalar Matrix:** Diagonal matrix with all diagonal elements equal
- **Identity Matrix:** Diagonal matrix with all diagonal elements 1
- **Zero Matrix:** Matrix with all elements zero

Key Formulas & Properties

- $A(adjA) = (adjA)A = |A|I$
- $(AB)^{-1} = B^{-1}A^{-1}$
- $(AB)^T = B^T A^T$
- $|adjA| = |A|^{n-1}$ where n is the order of matrix A
- If $|A| \neq 0$, then $A^{-1} = \frac{1}{|A|}adjA$
- For a square matrix A, A is:
 - Symmetric if $A^T = A$
 - Skew-symmetric if $A^T = -A$
 - Orthogonal if $AA^T = A^T A = I$

Determinant Properties

- $|AB| = |A||B|$
- $|A^T| = |A|$
- $|A^{-1}| = \frac{1}{|A|}$
- $|kA| = k^n|A|$ where n is order of A
- If any two rows/columns are identical, then $|A| = 0$
- If any row/column is all zeros, then $|A| = 0$

3 Matrices and Determinants - Practice Questions

Practice Questions (7)

1. If $A = \begin{bmatrix} 1 & 0 & 2 \\ 0 & 2 & 1 \\ 2 & 0 & 3 \end{bmatrix}$, then show that $A^3 - 6A^2 + 7A + 2I = 0$.

2. Express $A = \begin{bmatrix} 3 & -1 & 0 \\ 2 & 0 & 3 \\ 1 & -1 & 2 \end{bmatrix}$ as the sum of a symmetric and a skew-symmetric matrix.

3. If $A = \begin{bmatrix} 2 & 3 \\ -1 & 2 \end{bmatrix}$, then show that $A^2 - 4A + 7I = 0$. Hence find A^5 .

4. Express $\begin{bmatrix} 2 & 0 & 1 \\ 2 & 1 & 3 \\ 1 & -1 & 0 \end{bmatrix}$ as the sum of symmetric and skew symmetric matrices.

5. If $A = \begin{bmatrix} 2 & -3 & 5 \\ 3 & 2 & -4 \\ 1 & 1 & -2 \end{bmatrix}$, find A^{-1} and using A^{-1} solve the system: $2x - 3y + 5z = 11$, $3x + 2y - 4z = -5$,
 $x + y - 2z = -3$.

6. Use the product $\begin{bmatrix} 1 & -1 & 2 \\ 0 & 2 & -3 \\ 3 & -2 & 4 \end{bmatrix} \begin{bmatrix} -2 & 0 & 1 \\ 9 & 2 & -3 \\ 6 & 1 & -2 \end{bmatrix}$ and solve: $x + 3z = 9$, $-x + 2y - 2z = 4$,
 $2x - 3y + 4z = -3$.

7. Find the value of x and y, if $\begin{bmatrix} x + 3y & y \\ 7 - x & 4 \end{bmatrix} = \begin{bmatrix} 4 & -1 \\ 0 & 4 \end{bmatrix}$.

4 Continuity and Differentiability - Concepts & Formulas

Chapter Concepts

A function f is continuous at a point a if $\lim_{x \rightarrow a} f(x) = f(a)$. A function is differentiable at a point if its derivative exists at that point.

Continuity:

- A function is continuous at $x=a$ if:
 1. $f(a)$ exists
 2. $\lim_{x \rightarrow a} f(x)$ exists
 3. $\lim_{x \rightarrow a} f(x) = f(a)$
- Every differentiable function is continuous, but converse is not true

Differentiability:

- Left hand derivative: $f'(a^-) = \lim_{h \rightarrow 0^-} \frac{f(a+h) - f(a)}{h}$
- Right hand derivative: $f'(a^+) = \lim_{h \rightarrow 0^+} \frac{f(a+h) - f(a)}{h}$
- f is differentiable at $x=a$ if $f'(a^-) = f'(a^+)$

Key Formulas

- $\frac{d}{dx}(x^n) = nx^{n-1}$
- $\frac{d}{dx}(\sin x) = \cos x$
- $\frac{d}{dx}(\cos x) = -\sin x$
- $\frac{d}{dx}(\tan x) = \sec^2 x$
- $\frac{d}{dx}(\cot x) = -\operatorname{cosec}^2 x$
- $\frac{d}{dx}(\sec x) = \sec x \tan x$
- $\frac{d}{dx}(\operatorname{cosec} x) = -\operatorname{cosec} x \cot x$
- $\frac{d}{dx}(e^x) = e^x$
- $\frac{d}{dx}(a^x) = a^x \log a$
- $\frac{d}{dx}(\log x) = \frac{1}{x}$

Inverse Trigonometric Derivatives

- $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$

4 Continuity and Differentiability - Practice Questions

Practice Questions (8)

1. If $x^y = e^{x-y}$, then prove that $\frac{dy}{dx} = \frac{\log x}{(1+\log x)^2}$.
2. Differentiate the function $x^{\sin x} + (\sin x)^{\cos x}$ with respect to x .
3. If $x = \sqrt{a^{\sin^{-1} t}}$, $y = \sqrt{a^{\cos^{-1} t}}$, show that $\frac{dy}{dx} = -\frac{y}{x}$.
4. If $\sqrt{1-x^2} + \sqrt{1-y^2} = a(x-y)$ then show that $\frac{dy}{dx} = \sqrt{\frac{1-y^2}{1-x^2}}$.
5. If $y\sqrt{x^2+1} = \log(\sqrt{x^2+1}-x)$ prove that $(x^2+1)\frac{dy}{dx} + xy + 1 = 0$.
6. If $y = x^{\sin x} + (\sin x)^x$ find $\frac{dy}{dx}$.
7. If $y = \tan x + \sec x$, then prove that $\frac{d^2y}{dx^2} = \frac{\cos x}{(1-\sin x)^2}$.
8. For what value of k , is the function $f(x) = \begin{cases} \frac{\sqrt{4+x}-2}{x}, & x \neq 0 \\ k, & x = 0 \end{cases}$ continuous at $x=0$?

5 Application of Derivatives - Concepts & Formulas

Chapter Concepts

Derivatives have various applications including rate of change, increasing/decreasing functions, and finding maxima/minima.

Rate of Change:

- $\frac{dy}{dt}$ represents rate of change of y with respect to t
- If $y = f(x)$ and $x = g(t)$, then $\frac{dy}{dt} = \frac{dy}{dx} \cdot \frac{dx}{dt}$

Increasing/Decreasing Functions:

- Increasing function: $f'(x) > 0$
- Decreasing function: $f'(x) < 0$

Maxima and Minima

First Derivative Test:

- Find critical points where $f'(x) = 0$ or $f'(x)$ does not exist
- If $f'(x)$ changes from positive to negative at c , then $f(c)$ is local maximum
- If $f'(x)$ changes from negative to positive at c , then $f(c)$ is local minimum

Second Derivative Test:

- If $f'(c) = 0$ and $f''(c) < 0$, then $f(c)$ is local maximum
- If $f'(c) = 0$ and $f''(c) > 0$, then $f(c)$ is local minimum
- If $f''(c) = 0$, test fails

Practical Applications

- Optimization problems (maximizing area, volume, profit, etc.)
- Related rates problems
- Motion problems
- Economics applications (marginal cost, revenue, etc.)

5 Application of Derivatives - Practice Questions

Practice Questions (7)

1. Side of an equilateral triangle expands at the rate of 2 cm/sec. Find the rate of increase of its area when each side is 10 cm.

2. The radius r of the base of a right circular cone is decreasing at the rate of 2 cm/min and its height h is increasing at the rate of 3 cm/min. When $r=3.5$ cm and $h=6$ cm, find the rate of change of the volume of the cone. (Use $\pi = 22/7$)

3. The volume of a cube is increasing at the rate of $6 \text{ cm}^3/\text{s}$. How fast is the surface area of cube increasing, when the length of an edge is 8 cm?

4. A ladder 5 m long is leaning against a wall. Bottom of ladder is pulled along the ground away from wall at the rate of 2 m/s. How fast is the height on the wall decreasing, when the foot of ladder is 4 m away from the wall?

5. The radius of spherical balloon is increasing at the rate of 5cm/s. At what rate is the surface of the balloon increasing, when the radius is 10cm.

6. A rectangular visiting card is to contain 24 sq.cm. of printed matter. The margins at the top and bottom of the card are to be 1 cm and the margins on the left and right are to be $1\frac{1}{2}$ cm. (i) Write the expression for the area of the visiting card in terms of x . (ii) Obtain the dimensions of the card of minimum area.

7. A store has been selling calculators at ₹350 each. A market survey indicates that a reduction in price (p) of calculator increases the number of units (x) sold. The relation between the price and quantity sold is given by the demand function $p = 450 - \frac{1}{2}x$. (i) Determine the number of units (x) that should be sold to maximise the revenue $R(x) = xp(x)$. (ii) What rebate in price of calculator should the store give to maximise the revenue?

6 Integrals - Concepts & Formulas

Chapter Concepts

Integration is the inverse process of differentiation. It has applications in finding areas, volumes, and solving differential equations.

Indefinite Integrals:

- $\int f(x)dx = F(x) + C$ where $F'(x) = f(x)$
- C is the constant of integration

Definite Integrals:

- $\int_a^b f(x)dx = F(b) - F(a)$ where $F'(x) = f(x)$
- Definite integral represents the area under the curve

Basic Integration Formulas

- $\int x^n dx = \frac{x^{n+1}}{n+1} + C, n \neq -1$
- $\int \frac{1}{x} dx = \log|x| + C$
- $\int e^x dx = e^x + C$
- $\int a^x dx = \frac{a^x}{\log a} + C$
- $\int \sin x dx = -\cos x + C$
- $\int \cos x dx = \sin x + C$
- $\int \sec^2 x dx = \tan x + C$
- $\int \operatorname{cosec}^2 x dx = -\cot x + C$
- $\int \sec x \tan x dx = \sec x + C$
- $\int \operatorname{cosec} x \cot x dx = -\operatorname{cosec} x + C$

Integration Methods

- **Substitution:** $\int f(g(x))g'(x)dx = \int f(u)du$
- **Integration by parts:** $\int uvdx = u \int vdx - \int u'(\int vdx)dx$
- **Partial fractions:** For rational functions
- **Trigonometric substitutions:** For expressions with $\sqrt{a^2 - x^2}, \sqrt{a^2 + x^2}$, etc.

6 Integrals - Practice Questions

Practice Questions (8)

1. Find: $\int \frac{1}{x(x^2-1)} dx$.

2. Evaluate: $\int_0^{\pi/4} \frac{x dx}{1 + \cos 2x + \sin 2x}$

3. Find: $\int e^x \left[\frac{1}{(1+x^2)^{3/2}} + \frac{x}{\sqrt{1+x^2}} \right] dx$

4. Find: $\int \frac{3x+5}{\sqrt{x^2+2x+4}} dx$

5. Evaluate: $\int_0^3 \frac{dx}{\sqrt{9-x^2}}$

6. Find: $\int x^2 e^{x^3} dx$

7. Find: $\int \frac{1}{\sin^2 x \cos^2 x} dx$

8. Find: $\int \frac{\cos 2x - \cos 2\theta}{\cos x - \cos \theta} dx$

7 Application of Integrals - Concepts & Formulas

Chapter Concepts

Integrals are used to find areas under curves and between curves. The area bounded by the curve $y=f(x)$, x-axis and the ordinates $x=a$ and $x=b$ is given by $\int_a^b f(x) dx$.

Area Formulas:

- Area under curve: $\int_a^b f(x) dx$
- Area between two curves: $\int_a^b [f(x) - g(x)] dx$ where $f(x) \geq g(x)$
- Area bounded by curve $x=f(y)$ and y-axis: $\int_c^d f(y) dy$
- Area enclosed by ellipse: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$ is πab
- Area enclosed by circle: $x^2 + y^2 = a^2$ is πa^2

Area Calculation Methods

- **Vertical strips:** Area = $\int_a^b [f(x) - g(x)] dx$
- **Horizontal strips:** Area = $\int_c^d [f(y) - g(y)] dy$
- **Polar coordinates:** Area = $\frac{1}{2} \int_{\alpha}^{\beta} r^2 d\theta$
- **Parametric form:** Area = $\int_{t_1}^{t_2} y \frac{dx}{dt} dt$

Important Areas

- Area of triangle using coordinates: $\frac{1}{2} |x_1(y_2 - y_3) + x_2(y_3 - y_1) + x_3(y_1 - y_2)|$
- Area of sector: $\frac{1}{2} r^2 \theta$
- Area of segment: $\frac{1}{2} r^2 (\theta - \sin \theta)$

7 Application of Integrals - Practice Questions

Practice Questions (7)

1. Find the area bounded by the curve $xy = 4$, x-axis and the lines $x = 1$ and $x = 4$.

2. Find the area bounded by the curve $y = \cos x$ between 0 and 2π .

3. Find the area of the parabola bounded by $x^2 = 4y$, $y = 2$, $y = 4$ and y-axis in the first quadrant.

4. Find the area enclosed by the ellipse $\frac{x^2}{a^2} + \frac{y^2}{b^2} = 1$.

5. Find the area lying in the first quadrant and bounded by the circle $x^2 + y^2 = 4$ and the lines $x = 0$ and $x = 2$.

6. Find the area of the region enclosed by the parabola $y^2 = 4ax$ and the line $y = mx$.

7. Find the area of the region bounded by the parabola $y = x^2$ and $y = |x|$.

8 Differential Equations - Concepts & Formulas

Chapter Concepts

A differential equation is an equation involving derivatives of a function. The order is the order of the highest derivative, and the degree is the power of the highest order derivative.

Types of Differential Equations:

- **Variable separable:** $\frac{dy}{dx} = f(x)g(y)$
- **Homogeneous:** $\frac{dy}{dx} = f\left(\frac{y}{x}\right)$
- **Linear:** $\frac{dy}{dx} + P(x)y = Q(x)$
- **Exact:** $Mdx + Ndy = 0$ where $\frac{\partial M}{\partial y} = \frac{\partial N}{\partial x}$

Solution Methods

- **Variable separable:** $\int \frac{dy}{g(y)} = \int f(x)dx$
- **Homogeneous:** Substitute $y = vx$
- **Linear:** Integrating factor $e^{\int Pdx}$
- **Exact:** Find $F(x, y)$ such that $dF = Mdx + Ndy$

Important Differential Equations

- Exponential growth/decay: $\frac{dy}{dt} = ky$
- Newton's law of cooling: $\frac{dT}{dt} = -k(T - T_0)$
- Population growth: $\frac{dP}{dt} = kP\left(1 - \frac{P}{M}\right)$
- Simple harmonic motion: $\frac{d^2x}{dt^2} + \omega^2x = 0$

8 Differential Equations - Practice Questions

Practice Questions (8)

1. Find the particular solution of the differential equation: $\frac{dy}{dx} = y \cot 2x$, given that $y(\pi/4) = 2$.

2. Find the particular solution of the differential equation: $(xe^{y/x} + y)dx = xdy$, given that $y = 1$ when $x = 1$.

3. Solve the differential equation: $(1 + x^2)\frac{dy}{dx} + 2xy - 4x^2 = 0$, subject to the initial condition $y(0) = 0$.

4. Find the solution of the differential equation: $\log\left(\frac{dy}{dx}\right) = ax + by$.

5. Find the general solution of the differential equation: $\frac{dy}{dx} + \frac{1}{x} = \frac{e^y}{x}$.

6. Solve the differential equation: $(e^x + 1)ydy = e^x(y + 1)dx$.

7. Find the particular solution of the differential equation: $\frac{dy}{dx} = y \tan x$ when $y(0) = 1$.

8. Solve the differential equation: $x(x^2 - 1)\frac{dy}{dx} = 1$, $y = 0$ when $x = 2$.

9 Vectors - Concepts & Formulas

Chapter Concepts

Vectors have both magnitude and direction. They can be represented in component form or using unit vectors i , j , k .

Vector Operations:

- **Addition:** Triangle law, parallelogram law
- **Subtraction:** $\vec{a} - \vec{b} = \vec{a} + (-\vec{b})$
- **Scalar multiplication:** $k\vec{a}$ has magnitude $|k||\vec{a}|$
- **Dot product:** $\vec{a} \cdot \vec{b} = |\vec{a}||\vec{b}| \cos \theta$
- **Cross product:** $\vec{a} \times \vec{b} = |\vec{a}||\vec{b}| \sin \theta \hat{n}$

Key Formulas & Properties

- Dot product: $\vec{a} \cdot \vec{b} = a_x b_x + a_y b_y + a_z b_z$
- Cross product: $\vec{a} \times \vec{b} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ a_x & a_y & a_z \\ b_x & b_y & b_z \end{vmatrix}$
- Scalar projection of $\vec{a}$ on $\vec{b}$: $\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|}$
- Vector projection of $\vec{a}$ on $\vec{b}$: $\left(\frac{\vec{a} \cdot \vec{b}}{|\vec{b}|^2} \right) \vec{b}$
- Area of parallelogram = $|\vec{a} \times \vec{b}|$
- Area of triangle = $\frac{1}{2} |\vec{a} \times \vec{b}|$

9 Vectors - Practice Questions

Practice Questions (7)

1. Find a vector of magnitude 4 units perpendicular to each of the vectors $2\hat{i} - \hat{j} + \hat{k}$ and $\hat{i} + \hat{j} - \hat{k}$.

2. Find the position vector of point C which divides the line segment joining points A and B having position vectors $\hat{i} + 2\hat{j} - \hat{k}$ and $-\hat{i} + \hat{j} + \hat{k}$ respectively in the ratio 4:1 externally. Further, find $|AB| : |BC|$.

3. If vectors $\vec{a}$, $\vec{b}$ and $2\vec{a} + 3\vec{b}$ are unit vectors, then find the angle between $\vec{a}$ and $\vec{b}$.

4. If $\vec{a}$ and $\vec{b}$ are two non-zero vectors such that $(\vec{a} + \vec{b}) \perp \vec{a}$ and $(2\vec{a} + \vec{b}) \perp \vec{b}$, then prove that $|\vec{b}| = \sqrt{2}|\vec{a}|$.

5. In the given figure, ABCD is a parallelogram. If $\vec{AB} = 2\hat{i} - 4\hat{j} + 5\hat{k}$ and $\vec{DB} = 3\hat{i} - 6\hat{j} + 2\hat{k}$, then find $\vec{AD}$ and hence find the area of parallelogram ABCD.

6. If $\vec{a}$ and $\vec{b}$ are two vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 2$ and $\vec{a} \cdot \vec{b} = \sqrt{3}$, then find the angle between $2\vec{a}$ and $-\vec{b}$.

7. The vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - 3\hat{j} - 5\hat{k}$ and $\vec{c} = -3\hat{i} + 4\hat{j} + 4\hat{k}$ represent the sides of which type of triangle?

10 Three-Dimensional Geometry - Concepts & Formulas

Chapter Concepts

3D geometry deals with points, lines, and planes in three-dimensional space. Direction cosines and ratios are used to represent the direction of lines.

Direction Cosines and Ratios:

- Direction cosines: $l = \cos \alpha, m = \cos \beta, n = \cos \gamma$
- $l^2 + m^2 + n^2 = 1$
- Direction ratios: a, b, c proportional to l, m, n

Line Equations:

- Vector form: $\vec{r} = \vec{a} + \lambda \vec{b}$
- Cartesian form: $\frac{x-x_1}{a} = \frac{y-y_1}{b} = \frac{z-z_1}{c}$

Key Formulas

- Distance between points: $\sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2 + (z_2 - z_1)^2}$
- Angle between lines: $\cos \theta = \frac{|a_1 a_2 + b_1 b_2 + c_1 c_2|}{\sqrt{a_1^2 + b_1^2 + c_1^2} \sqrt{a_2^2 + b_2^2 + c_2^2}}$
- Shortest distance between skew lines: $\frac{|(\vec{b}_1 \times \vec{b}_2) \cdot (\vec{a}_2 - \vec{a}_1)|}{|\vec{b}_1 \times \vec{b}_2|}$
- Distance from point to line: $\frac{|\vec{AB} \times \vec{b}|}{|\vec{b}|}$
- Distance between parallel lines: $\frac{|\vec{AB} \times \vec{b}|}{|\vec{b}|}$

10 Three-Dimensional Geometry - Practice Questions

Practice Questions (8)

1. Find the vector and Cartesian equations of a line passing through $(1, 2, -4)$ and parallel to the line joining $A(3, 3, -5)$ and $B(1, 0, -11)$. Hence, find the distance between the two lines.
2. Find the equations of the line passing through $A(1, 2, 3)$ and $B(3, 5, 9)$. Hence, find coordinates of points on this line which are at a distance of 14 units from B.
3. A line l passes through $(-1, 3, -2)$ and is perpendicular to both lines $\frac{x}{1} = \frac{y}{2} = \frac{z}{3}$ and $\frac{x+2}{-3} = \frac{y-1}{2} = \frac{z+1}{5}$. Find the vector equation of l and its distance from origin.
4. Find the equations of the diagonals of parallelogram PQRS with vertices $P(4, 2, -6)$, $Q(5, -3, 1)$, $R(12, 4, 5)$, $S(11, 9, -2)$. Use these to find intersection of diagonals.
5. The image of point $P(x, y, z)$ with respect to line $\frac{x}{1} = \frac{y-1}{2} = \frac{z-2}{3}$ is $P'(1, 0, 7)$. Find coordinates of P.
6. Find coordinates of image of $(1, 6, 3)$ with respect to line $\vec{r} = (\hat{j} + 2\hat{k}) + \lambda(\hat{i} + 2\hat{j} + 3\hat{k})$. Also find distance of image from y-axis.
7. Find the shortest distance between lines $\frac{x-8}{3} = \frac{y+9}{-16} = \frac{z-10}{7}$ and $\frac{x-15}{3} = \frac{y-29}{8} = \frac{z-5}{-5}$.
8. Find the value of p for which lines $\vec{r} = \lambda\hat{i} + (2\lambda + 1)\hat{j} + (3\lambda + 2)\hat{k}$ and $\vec{r} = \hat{i} - 3\mu\hat{j} + (p\mu + 7)\hat{k}$ are perpendicular and intersect. Also find point of intersection.

11 Linear Programming - Concepts & Formulas

Chapter Concepts

Linear programming involves optimizing a linear objective function subject to linear constraints. The graphical method is used for problems with two variables.

Key Terms:

- **Objective function:** Function to be maximized/minimized
- **Constraints:** Linear inequalities representing restrictions
- **Feasible region:** Region satisfying all constraints
- **Optimal solution:** Point in feasible region optimizing objective function
- **Corner point method:** Evaluate objective function at all corner points

Solution Methods

- **Graphical method:** For problems with 2 variables
 1. Plot constraints to find feasible region
 2. Find corner points of feasible region
 3. Evaluate objective function at each corner point
 4. Optimal solution is at a corner point
- **Simplex method:** For problems with more variables

Important Results

- Feasible region is always a convex polygon
- Optimal solution always occurs at a corner point
- If feasible region is unbounded, optimal solution may not exist
- If two corner points give same optimal value, all points on the line segment between them are optimal

11 Linear Programming - Practice Questions

Practice Questions (7)

1. Solve graphically: Maximize $Z = x + 2y$ subject to $x + 2y \geq 100$, $2x - y \leq 0$, $2x + y \leq 200$, $x \geq 0$, $y \geq 0$.

2. Solve graphically: Minimize $z = x + 2y$ subject to $2x + y \geq 3$, $x + 2y \geq 6$, $x \geq 0$, $y \geq 0$.

3. Solve graphically: Minimize $Z = 200x + 500y$ subject to $x + 2y \geq 10$, $3x + 4y \leq 24$, $x \geq 0$, $y \geq 0$.

4. Solve graphically: Minimize $z = -3x + 4y$ subject to $x + 2y \leq 8$, $3x + 2y \leq 12$, $x, y \geq 0$.

5. Solve graphically: Minimize $z = x + 2y$ subject to $x + 2y \geq 100$, $2x - y \leq 0$, $2x + y \leq 200$, $x, y \geq 0$.

6. Solve graphically: Minimize $z = 3x + 8y$ subject to $3x + 4y \geq 8$, $5x + 2y \geq 11$, $x \geq 0$, $y \geq 0$.

7. Solve graphically: Maximize $Z = 2x + 3y$ subject to $x + y \leq 6$, $x \geq 2$, $y \leq 3$, $x, y \geq 0$.

12 Probability - Concepts & Formulas

Chapter Concepts

Probability measures the likelihood of events. Conditional probability, Bayes' theorem, and probability distributions are important concepts.

Basic Concepts:

- Sample space: Set of all possible outcomes
- Event: Subset of sample space
- Probability: $P(E) = \frac{\text{Number of favorable outcomes}}{\text{Total number of outcomes}}$
- Complement: $P(E') = 1 - P(E)$
- Union: $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
- Intersection: $P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$

Key Formulas

- Conditional probability: $P(A|B) = \frac{P(A \cap B)}{P(B)}$
- Multiplication theorem: $P(A \cap B) = P(A)P(B|A) = P(B)P(A|B)$
- Bayes' theorem: $P(E_i|A) = \frac{P(E_i)P(A|E_i)}{\sum P(E_j)P(A|E_j)}$
- Mean of random variable: $E(X) = \sum x_i P(x_i)$
- Variance: $Var(X) = E(X^2) - [E(X)]^2$
- Standard deviation: $\sigma = \sqrt{Var(X)}$

12 Probability - Practice Questions

Practice Questions (8)

1. Out of two bags, bag A contains 2 white and 3 red balls and bag B contains 4 white and 5 red balls. One ball is drawn at random from one of the bags and is found to be red. Find the probability that it was drawn from bag B.
2. A card from a well shuffled deck of 52 playing cards is lost. From the remaining cards of the pack, a card is drawn at random and is found to be a King. Find the probability of the lost card being a King.
3. A departmental store sends bills to charge its customers once a month. Past experience shows that 70% of its customers pay their first month bill in time. The store also found that the customer who pays the bill in time has the probability of 0.8 of paying in time next month and the customer who doesn't pay in time has the probability of 0.4 of paying in time the next month. (i) Let E_1 and E_2 respectively denote the event of customer paying or not paying the first month bill in time. Find $P(E_1)$, $P(E_2)$. (ii) Let A denotes the event of customer paying second month's bill in time, then find $P(A|E_1)$ and $P(A|E_2)$. (iii) Find the probability of customer paying second month's bill in time.
4. A building contractor undertakes a job to construct 4 flats on a plot along with parking area. Due to strike the probability of many construction workers not being present for the job is 0.65. The probability that many are not present and still the work gets completed on time is 0.35. The probability that work will be completed on time when all workers are present is 0.80. Let E_1 : represent the event when many workers were not present for the job; E_2 : represent the event when all workers were present; and E: represent completing the construction work on time. (i) What is the probability that all the workers are present for the job? (ii) What is the probability that construction will be completed on time? (iii) What is the probability that many workers are not present given that the construction work is completed on time?
5. Previous records state that the probability of an airplane crash is 0.00001%. Further, there are 95% chances that there will be survivors after a plane crash. Assume that in case of no crash, all travellers survive. Let E_1 be the event that there is a plane crash and E_2 be the event that there is no crash. Let A be the event that passengers survive after the journey. (i) Find the probability that the airplane will not crash. (ii) Find $P(A|E_1) + P(A|E_2)$. (iii) Find $P(A)$.
6. Recent studies suggest that roughly 12% of the world population is left handed. Depending upon the parents, the chances of having a left handed child are as follows: A: When both father and mother are left handed: Chances of left handed child is 24%. B: When father is right handed and mother is left handed: Chances of left handed child is 22%. C: When father is left handed and mother is right handed: Chances of left handed child is 17%. D: When both father and mother are right handed: Chances of left handed child is 9%. Assuming that $P(A) = P(B) = P(C) = P(D) = 1/4$ and L denotes the event that child is left handed. (i) Find $P(L|C)$ (ii) Find $P(\text{not } L|A)$ (iii) Find $P(A|L)$
7. If A and B are two events such that $P(A) = 0.2$, $P(B) = 0.4$ and $P(A \cup B) = 0.6$, find $P(A|B)$.
8. If $P(A \cap B) = 1/2$ and $P(A' \cap B') = 1/3$, $P(A) = P$, $P(B) = 2P(A)$, find the value of P.

13 Mixed Questions Set 1 - Concepts & Formulas

Revision Strategy

This day focuses on solving mixed problems from various chapters to enhance problem-solving skills and application of concepts.

Revision Tips:

- Focus on understanding concepts rather than memorization
- Practice previous year questions
- Revise all formulas and theorems
- Solve mixed problems to integrate concepts
- Time yourself while solving problems

Important Formulas to Remember

- **Relations:** Reflexive, symmetric, transitive properties
- **Functions:** One-one, onto, composition, inverse
- **Matrices:** Operations, determinants, inverse
- **Differentiation:** All standard formulas and rules
- **Integration:** Basic formulas and methods
- **Vectors:** Dot product, cross product, projection
- **3D Geometry:** Direction ratios, lines, planes
- **Probability:** Conditional probability, Bayes theorem

Problem Solving Strategies

- Read questions carefully
- Identify the concept being tested
- Show all steps in solutions
- Check your answers
- Manage time effectively

13 Mixed Questions Set 1 - Practice Questions

Practice Questions (8)

1. If the matrix $\begin{bmatrix} a & c & 0 \\ b & d & 0 \\ 0 & 0 & 5 \end{bmatrix}$ is a scalar matrix, find the value of $a + 2b + 3c + 4d$.

2. Given that $A^{-1} = \frac{1}{7} \begin{bmatrix} 2 & 1 \\ -3 & 2 \end{bmatrix}$, find matrix A.

3. If $A = \begin{bmatrix} 2 & 1 \\ -4 & -2 \end{bmatrix}$, find the value of $I - A + A^2 - A^3 + \dots$.

4. Find the general solution of the differential equation $xdy + ydx = 0$.

5. Find the integrating factor of $(x + 2y^2) \frac{dy}{dx} = y$ ($y > 0$).

6. The vectors $\vec{a} = 2\hat{i} - \hat{j} + \hat{k}$, $\vec{b} = \hat{i} - 3\hat{j} - 5\hat{k}$ and $\vec{c} = -3\hat{i} + 4\hat{j} + 4\hat{k}$ represent the sides of which type of triangle?

7. Let $\vec{a}$ be any vector such that $|\vec{a}| = a$. Find the value of $|\vec{a} \times \hat{i}|^2 + |\vec{a} \times \hat{j}|^2 + |\vec{a} \times \hat{k}|^2$.

8. The lines $\frac{1-x}{2} = \frac{y-1}{3} = \frac{z}{1}$ and $\frac{2x-3}{2p} = \frac{y}{-1} = \frac{z-4}{7}$ are perpendicular for what value of p?

14 Mixed Questions Set 2 - Concepts & Formulas

Advanced Problem Solving

This day focuses on more challenging problems that require integration of concepts from multiple chapters.

Problem Solving Strategies:

- Break down complex problems into smaller parts
- Identify the key concepts being tested
- Use multiple approaches to solve problems
- Check your answers for reasonableness
- Practice time management

Important Integration Formulas

- $\int \frac{dx}{x^2+a^2} = \frac{1}{a} \tan^{-1} \frac{x}{a} + C$
- $\int \frac{dx}{x^2-a^2} = \frac{1}{2a} \log \left| \frac{x-a}{x+a} \right| + C$
- $\int \frac{dx}{\sqrt{a^2-x^2}} = \sin^{-1} \frac{x}{a} + C$
- $\int \frac{dx}{\sqrt{x^2+a^2}} = \log |x + \sqrt{x^2+a^2}| + C$
- $\int \sqrt{a^2-x^2} dx = \frac{x}{2} \sqrt{a^2-x^2} + \frac{a^2}{2} \sin^{-1} \frac{x}{a} + C$
- $\int \sqrt{x^2+a^2} dx = \frac{x}{2} \sqrt{x^2+a^2} + \frac{a^2}{2} \log |x + \sqrt{x^2+a^2}| + C$

Important Differentiation Formulas

- $\frac{d}{dx}(u^v) = u^v \left(\frac{v}{u} \frac{du}{dx} + \log u \frac{dv}{dx} \right)$
- $\frac{d}{dx}(\sin^{-1} x) = \frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\cos^{-1} x) = -\frac{1}{\sqrt{1-x^2}}$
- $\frac{d}{dx}(\tan^{-1} x) = \frac{1}{1+x^2}$
- $\frac{d}{dx}(\cot^{-1} x) = -\frac{1}{1+x^2}$
- $\frac{d}{dx}(\sec^{-1} x) = \frac{1}{|x|\sqrt{x^2-1}}$
- $\frac{d}{dx}(\operatorname{cosec}^{-1} x) = -\frac{1}{|x|\sqrt{x^2-1}}$

14 Mixed Questions Set 2 - Practice Questions

Practice Questions (8)

1. Given that $y = (\sin x)^x \cdot x^{\sin x} + a^x$, find $\frac{dy}{dx}$.

2. Evaluate: $\int_0^{\pi/4} \frac{x dx}{1 + \cos 2x + \sin 2x}$

3. Find the particular solution of $\frac{dy}{dx} = y \cot 2x$, given that $y(\pi/4) = 2$.

4. Solve the LPP graphically: Maximize $Z = 2x + 3y$ subject to $x + y \leq 6, x \geq 2, y \leq 3, x, y \geq 0$.

5. A card from a well shuffled deck of 52 cards is lost. From the remaining cards, a card is drawn and found to be a King. Find the probability of the lost card being a King.

6. Sketch the graph of $y = x|x|$ and find the area bounded by this curve, X-axis and $x = -2, x = 2$.

7. Let $A = \mathbb{R} - \{5\}$ and $B = \mathbb{R} - \{1\}$. Consider $f : A \rightarrow B$ defined by $f(x) = \frac{x-3}{x-5}$. Show that f is one-one and onto.

8. If $A = \begin{bmatrix} 2 & 1 & -3 \\ 3 & 2 & 1 \\ 1 & 2 & -1 \end{bmatrix}$, find A^{-1} and solve: $2x + y - 3z = 13, 3x + 2y + z = 4, x + 2y - z = 8$.

15 Mixed Questions Set 3 & Revision - Concepts & Formulas

Final Revision Strategy

This day focuses on application-based problems and final revision of all concepts covered in the syllabus.

Revision Tips:

- Focus on understanding concepts rather than memorization
- Practice previous year questions
- Revise all formulas and theorems
- Solve mixed problems to integrate concepts
- Time yourself while solving problems

Important Formulas to Remember

- **Relations:** Reflexive, symmetric, transitive properties
- **Functions:** One-one, onto, composition, inverse
- **Matrices:** Operations, determinants, inverse
- **Differentiation:** All standard formulas and rules
- **Integration:** Basic formulas and methods
- **Vectors:** Dot product, cross product, projection
- **3D Geometry:** Direction ratios, lines, planes
- **Probability:** Conditional probability, Bayes theorem

Exam Strategy

- Read questions carefully
- Start with questions you're most confident about
- Show all steps in solutions
- Manage time effectively
- Review answers if time permits

15 Mixed Questions Set 3 & Revision - Practice Questions

Practice Questions (10)

1. A store sells calculators at ₹350 each. The demand function is $p = 450 - \frac{x}{2}$. Determine: (i) Number of units to maximize revenue $R(x) = xp(x)$ (ii) Rebate in price to maximize revenue.
2. An instructor at the astronomical centre shows three stars with position vectors: $\vec{D} = 2\hat{i} + 3\hat{j} + 4\hat{k}$, $\vec{A} = 7\hat{i} + 5\hat{j} + 8\hat{k}$, $\vec{V} = -3\hat{i} + 7\hat{j} + 11\hat{k}$. Find: (i) Distance between V and A (ii) Unit vector in direction of $\vec{DA}$ (iii) Measure of $\angle VDA$.
3. Rohit, Jaspreet and Alia appeared for an interview for three vacancies in the same post. The probability of Rohit's selection is $\frac{1}{5}$, Jaspreet's selection is $\frac{1}{3}$ and Alia's selection is $\frac{1}{4}$. The event of selection is independent of each other. Find: (i) $P(\text{at least one selected})$ (ii) $P(\text{Jaspreet selected} | \text{Rohit not selected})$ (iii) $P(\text{exactly one selected})$.
4. Check whether the relation S in $\mathbb{R}$ defined by $S = \{(a, b) : \text{where } a - b + \sqrt{2} \text{ is irrational}\}$ is reflexive, symmetric or transitive.
5. Using integration, find the area bounded by the ellipse $9x^2 + 25y^2 = 225$, the lines $x = -2$, $x = 2$, and the X-axis.
6. A salesman receives commission with daily income $Y = 800X + 50$, where X is number of sales with distribution: $X: 0, 1, 2, 3, 4, 5$; $P(X): 0.42, 0.3, 0.05, 0.2, 0.03$. Find: (i) k (ii) $P(X \geq 3)$ (iii) Expected weekly income for 5-day week.
7. Show that: $\tan\left(\frac{1}{2}\sin^{-1}\left(\frac{3}{4}\right)\right) = \frac{4-\sqrt{7}}{3}$.
8. Evaluate: $\cos^{-1}(\cos(\frac{2\pi}{3})) + \sin^{-1}(\sin(\frac{2\pi}{3}))$.
9. Write the value of $\tan^{-1}[2\sin(2\cos^{-1}(\frac{\sqrt{3}}{2}))]$.
10. Find the value of $\sin[\frac{\pi}{3} - \sin^{-1}(-\frac{1}{2})]$.